

### *Eligibility Criteria*

Eligibility criteria were defined according to the population, intervention, comparator, outcomes, and study design framework. Studies involving adults with T2DM were eligible if they evaluated an SGLT2 inhibitor, a GLP-1 receptor agonist, or the use of both classes and reported relevant cardiovascular, heart failure, kidney, mortality, metabolic, or safety outcomes. Eligible comparators included placebo or usual care, active comparators, the alternative drug class, and concomitant use of the other class.

Eligible study designs included randomized controlled trials, systematic reviews and meta-analyses, and large comparative observational studies. Studies were excluded if they focused solely on glycemic outcomes without reporting relevant clinical outcomes, involved pediatric populations or preclinical models, were case reports or editorials, represented duplicate or overlapping reports without unique data, or did not provide adequate information for extraction. When multiple reports described the same underlying study population, reports were linked to avoid double-counting.

Clinical guidelines, consensus statements, narrative reviews, cost-effectiveness analyses, and descriptive or survey studies cited elsewhere in the manuscript for background, clinical context, implementation, cost, or access were not considered part of the systematic evidence synthesis.

No statistical sample-size calculation was performed because all studies meeting the prespecified eligibility criteria were considered for inclusion. The final number of included studies was therefore determined by the systematic search and screening process rather than by a predetermined sample size.

### *Information Sources and Search Period*

Systematic searches were conducted in PubMed/MEDLINE, Embase, and Cochrane CENTRAL. Google Scholar was used as an additional source, and backward and forward citation searching was performed to identify additional potentially eligible studies. Searches were run on August 30, 2026, and covered studies published from January 1, 2010, through August 30, 2026.

January 1, 2010, was selected as the starting date to capture the contemporary evidence base leading to and following the major cardiovascular and cardiorenal outcome programs for SGLT2 inhibitors and GLP-1 receptor agonists. The search period was intended to encompass the development of clinically relevant cardiovascular, heart failure, kidney, metabolic, and safety evidence for these classes rather than the broader historical literature on glucose-lowering therapies.

### *Search Strategy*

Three complementary searches were conducted within each primary bibliographic database: an SGLT2 inhibitor class-specific search, a GLP-1 receptor agonist class-specific search, and a direct comparison or combination search. Separate class-specific searches were used so that potentially eligible studies were not required to mention both drug classes.

Search terminology incorporated population terms, alternative class terminology, individual drug names, and relevant clinical outcomes. Population terms included "type 2 diabetes," "type 2 diabetes mellitus," and "T2DM." SGLT2 inhibitor terms included "SGLT2," "SGLT-2," "SGLT2 inhibitor," "SGLT2 inhibitors," "sodium-glucose cotransporter 2 inhibitor," "sodium-glucose cotransporter 2 inhibitors," "empagliflozin, canagliflozin, dapagliflozin, and ertugliflozin. GLP-1 receptor agonist terms included "GLP-1," "GLP1," "GLP-1 receptor agonist," "GLP-1 receptor agonists," "glucagon-like peptide-1 receptor agonist," "glucagon-like peptide-1 receptor agonists," "liraglutide, semaglutide, dulaglutide, exenatide, lixisenatide, and albiglutide. Outcome terms included cardiovascular disease, major adverse cardiovascular events (MACE), atherosclerotic cardiovascular disease (ASCVD), heart failure, heart failure hospitalization, stroke, mortality, cardiovascular death, renal and kidney outcomes, CKD, estimated glomerular filtration rate (eGFR), kidney failure, kidney disease progression, safety, and adverse events. Direct comparison and combination searches additionally included HbA1c and weight.

In PubMed/MEDLINE, search terms were restricted to Title/Abstract fields. Corresponding searches in Embase and Cochrane CENTRAL were adapted to their respective title and abstract syntax. The complete database-specific search strategies, including the exact Boolean search strings, search terms, field restrictions, search dates, date limits, and retrieval counts for each search, are provided in Appendix 2.

The database searches identified 15,774 records in PubMed/MEDLINE, 27,908 in Embase, and 2,587 in Cochrane CENTRAL, yielding 46,269 database records before deduplication. Google Scholar was searched separately for SGLT2 inhibitors, GLP-1 receptor agonists, and direct comparison or combination evidence. Because of the large number of Google Scholar results, the first 200 results from each of the three searches were screened, for a total of 600 Google Scholar records. Backward and forward citation searching identified an additional 47 records.